

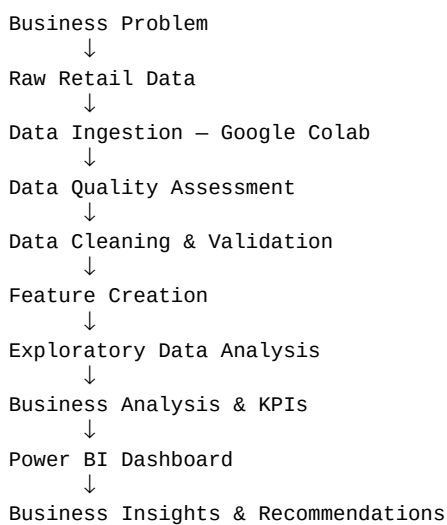
Project Deliverables

End-to-End Data Analysis Project — Google Colab + Power BI

1. Core Project Deliverable

Indian_Retail_Ecommerce_Sales_Customer_Product_Analysis is an **end-to-end data analysis project** developed using Google Colab for data analysis and Power BI for interactive dashboard reporting. The project covers the complete journey from raw retail data to cleaned data, exploratory analysis, business analysis, KPIs, dashboard reporting and business insights.

2. End-to-End Workflow



3. Data Analysis Deliverables

- Raw dataset import and inspection
- Missing-value analysis and treatment
- Duplicate-row and duplicate Order ID analysis
- Numeric and categorical validation
- Data standardization and consistency checks
- IQR-based outlier investigation
- Feature creation including Month, Year and Age_Group
- Exploratory Data Analysis
- Business-focused sales, customer, product, regional and operational analysis
- KPI calculation and business insights

4. Power BI Dashboard Deliverable

The project includes a Power BI dashboard/template as the visualization and reporting layer. The Power BI file should be stored separately in the **powerbi/** folder.

```
powerbi/  
■■■ Indian_Retail_Ecommerce_Sales_Customer_Product_Analysis_Dashboard.pbix
```

The dashboard contains the project reporting layer for presenting the analyzed data and business KPIs in an interactive format.

5. Data Output Deliverable

The notebook exports **cleaned_sales_data.csv**. The final cleaned core dataset contains **100,000 rows × 19 original columns**, with derived analytical fields such as Month, Year and Age_Group created during the analysis.

6. Key Analysis Areas

- Sales and revenue analysis
- Category and product performance
- State and city performance
- Customer segment analysis
- Age and gender analysis
- Online vs Store channel analysis
- Payment-method analysis
- Delivery-status analysis
- Monthly and yearly trends
- Discount, rating, quantity and age relationships
- Correlation analysis

7. Key KPI Deliverables

The project calculates Total Sales, Total Orders, Average Sales Amount and Total Quantity Sold, and identifies leading category, state, city, sales channel, customer segment, payment method and delivery status results.

8. Recommended GitHub Project Structure

```
Indian_Retail_Ecommerce_Sales_Customer_Product_Analysis/  
├──  
│   ├── notebooks/  
│   │   └── Indian_Retail_Ecommerce_Sales_Customer_Product_Analysis.ipynb  
│   ├── data/  
│   │   ├── processed/  
│   │   │   └── cleaned_sales_data.csv  
│   ├── powerbi/  
│   │   └── Indian_Retail_Ecommerce_Sales_Customer_Product_Analysis_Dashboard.pbix  
│   ├── docs/  
│   │   ├── Indian_Retail_Ecommerce_Sales_Customer_Product_Analysis_Business_Problem_Statement.pdf  
│   │   └── Indian_Retail_Ecommerce_Sales_Customer_Product_Analysis_Deliverables.pdf  
│   ├── README.md  
│   ├── .gitignore  
│   └── LICENSE
```

9. Documentation Deliverables

- Business Problem Statement
- Project Deliverables document
- README documentation
- Project structure and setup instructions

10. Project Completion Criteria

The project is complete when the data-analysis workflow is reproducible in Google Colab, the cleaned dataset is generated successfully, the major business questions are analyzed, KPIs and insights are documented, and the Power BI dashboard/template is included in the repository.

11. Scope Note

This project is presented as an **end-to-end data analysis project**. Google Colab/Python are used for data preparation and analysis, while Power BI is used as the dashboard/reporting layer. SQL is not listed as a completed deliverable because it is not part of the supplied code file.